

Methodological Review

1. Summary and overall assessment

The manuscript measures the retrieval-stage penalty that Moroccan Darija queries incur against an MSA knowledge base (Recall@1 0.706 → 0.559, 95% CI on the gap [0.090, 0.185]), shows that query normalisation does not repair it and offers a mechanistic account (margin collapse), and reports that cross-encoder reranking recovers a large part of the loss. Four of six findings are negative or diagnostic, and every headline carries a bootstrap interval. That honesty, together with a released corpus and a parallel benchmark, is the paper's real strength and clearly separates it from the extractive-QA framing of Althobaiti's ArDQA work.

Recommendation: major revision. The retrieval-stage results are publishable now. The end-to-end generation results are not, for reasons that are structural rather than cosmetic: the correctness metric rests on an unvalidated instrument, and the pipeline's outputs are shown below to be sensitive to an undocumented numerical setting. Both are fixable, and both will be the first things an MDPI reviewer asks about.

2. Originality of the topic and of each contribution

The paper's originality is uneven across its contributions, and it should be claimed unevenly. What is genuinely new is the object of measurement: a retrieval-stage dialect penalty, quantified with a parallel MSA/Darija benchmark over a real 3,054-passage corpus. The existing Darija resources (DarijaBERT, DODa, Atlas-Chat and its DarijaBench, Shang et al. 2024) evaluate understanding and generation, not retrieval, and ArDQA is extractive. No parallel dialect benchmark for retrieval is known to this reviewer, so this is the defensible claim and it should lead the Introduction. The second novel result, that the penalty concentrates at rank 1 and is absorbed by a top-5 generator, is under-argued: it sits in tension with Liu et al. 2023 (position within the context matters) and Cuconasu et al. 2024 (distractors matter), and the flat k-sweep contradicts the latter directly. The authors should discuss why this setting differs rather than leave the tension unaddressed.

The remaining contributions are partially novel or incremental. The finding that query normalisation hurts is new for Darija, but the mechanism is the classic cross-lingual IR query-translation ambiguity effect (Ballesteros & Croft 1997): a fluent translation gains overlap with the target and loses the discriminative terms that decide rank. The NADI 2023 and 2024 dialect-to-MSA machine translation subtasks are the direct precedent for the normaliser and are uncited; the honest framing is that effect reproduced at the dialect level, with the margin diagnostic as the added value. Cross-encoder reranking is standard two-stage retrieval applied to a new mismatch; the dialect-asymmetry claim would have made it a contribution, and it did not survive its own paired test (issue E below). Margin-gated selective reranking descends from confidence-gated cascades (Wang, Lin & Metzler 2011) and uses a known gate; its only new element, dialect-awareness, is checked rather than established, so it should be presented as engineering.

Two findings rediscover known effects. The author-style confound in benchmark construction is the annotation-artifact phenomenon (Gururangan et al. 2018; Geva, Goldberg & Berant 2019); it is a valuable caution for Arabic benchmark builders but the lineage must be cited or a reviewer will read the paper as unaware of it. Generator behaviour on retrieval failure is exactly the question NoMIRACL (Thakur et al. 2024) studies across 18 languages including Arabic; the $n = 7$ analysis here adds nothing and should either cite that work and be dropped, or be scaled. Finally, LLM-as-judge evaluation is standard (Zheng et al. 2023) but the specific configuration used, a judge from the same model family as the generator, is the one Panickssery, Bowman & Feng 2024 show to carry self-preference bias.

Originality verdict. The topic is original at the level of what is measured and what is released, not at the level of method, and three findings rediscover known effects without citing them. For a Q1 venue the Introduction must claim the first two contributions explicitly and position the rest against the works above; a reviewer who knows the cross-lingual IR or annotation-artifact literature will otherwise read the paper as overclaiming, which is a harder objection to recover from than any statistical one.

3. Major methodological issues that must be addressed

- A. The LLM-as-judge is unvalidated, and it judges its own model family.** Every correctness and faithfulness figure is a verdict from Qwen2.5-7B on answers written by Qwen2.5-7B. No Cohen's kappa against a human labeller is reported; the judge-validation notebook exists but its 100-row sheet has not been labelled. Reviewers will not accept a same-family judge without human agreement (target kappa of at least 0.6, with raw agreement alongside) and at least one independent judge. Until then, 'correctness' is a proxy of unknown validity.
- B. The faithfulness metric conflates refusal with unfaithfulness.** A refusal ('the information is not available in the texts') is not 'supported by the reference texts', so the judge scores refusals near-randomly: in the committed run, 19 of 35 refusals were scored faithful and 16 unfaithful on identical behaviour. Report faithfulness both including and excluding refusals, and treat refusal as its own outcome.
- C. Generation results are sensitive to an undocumented dtype and are not reproducible as written.** Two runs of the same pipeline on the same 200 items, same prompt and same greedy decoding differ only in whether the non-quantised modules of the 4-bit model run in float16 or float32. Retrieval was identical (gold-in-context agreed on 800 of 800 rows), yet 92% of generated answers differed and correctness moved from 0.880 to 0.725, with the oracle condition falling from 0.960 to 0.730. The Methods must state the quantisation configuration, compute dtype, library versions, decoding parameters, seed and hardware, and should include a dtype sensitivity note.
- D. Three inconsistent sets of generation numbers circulate in the repository.** The findings summary reports $n = 68$ (0.926 vs 0.882); the committed $n = 200$ run reports 0.885 vs 0.830; a second $n = 200$ run reports 0.795 vs 0.655. The manuscript must commit to one canonical run, name the file it came from, and archive it with a DOI. A table whose numbers cannot be traced to a single artefact is a desk-reject risk.
- E. The paper's novel claim, that reranking helps Darija more than MSA, is not demonstrated.** Two separate confidence intervals do not establish that two gains differ. The paired difference-of-differences bootstrap gives, for the best reranker, a difference of +0.060 with 95% CI [-0.0001, +0.120]; all four rerankers return no significant asymmetry. Soften the claim to a descriptive observation, or enlarge the sample until the paired test resolves it.
- F. No correction for multiple comparisons.** The mitigation grid alone is four encoders by four strategies by three metrics; the reranker study adds four models by three depths by two query types. Well over fifty 95% intervals are reported with no Holm or Benjamini-Hochberg adjustment and no pre-declared primary hypothesis. Declare the primary comparison (MSA versus Darija Recall@1), correct the secondary ones, and say so in the Methods.
- G. Prompt sensitivity is reported as a finding but not handled as a nuisance factor.** One added instruction moved dialectal correctness from 0.824 to 0.882 and flipped the MSA-Darija gap from significant (+0.132) to not significant (+0.044), so the central end-to-end conclusion is conditional on a single prompt. Run at least three semantically equivalent prompts and report the spread; the claim that a RAG pipeline absorbs the penalty needs to survive that.

- H. Hyper-parameter selection and evaluation share the same 200 items.** The hybrid weight, rerank depth and generator context size were chosen on the data on which results are reported. State a held-out development split or use nested cross-validation; otherwise every reported gap is optimistically biased by an unknown amount.
- I. Sub-analyses are underpowered, and null results are presented as if they were negative results.** Retrieval-failure behaviour rests on $n = 7$ and two runs that disagree (refusal 0.714 versus 0.143). The k-sweep is flat (0.835 to 0.845) with every interval spanning zero, which cannot separate 'context size does not matter' from 'undetectable at this n '. Add a power statement: the minimum detectable paired difference at $n = 200$ is roughly 0.06.
- J. Benchmark validity needs inter-annotator evidence.** Items were author-written or AI-assisted, then validated by a lexicon check and an LLM judge. The author-style confound is well documented, but Darija authenticity has no agreement between independent native speakers. Two annotators on a 100-item sample with kappa is the minimum.
- K. Generalisation is asserted from one dialect, one domain and one generator.** A second generator (Atlas-Chat is already named in the configuration) is the minimum before 'a RAG pipeline absorbs the penalty' can be stated as a property of RAG rather than of Qwen2.5-7B; a second dialect would strengthen external validity.

4. What MDPI (Q1) reviewers and editors will expect

On novelty, an explicit delimitation from ArDQA, from the Darija LLM benchmarks, from cross-lingual IR query translation and from NoMIRACL; at present only ArDQA is addressed. On reproducibility, pinned library versions, seeds, the exact prompts, decoding parameters, hardware and dtype, with code and data under a persistent DOI; the notebooks were Colab-only with hard-coded paths, now corrected, and the dtype remains undocumented. On statistics, effect sizes with confidence intervals, paired tests wherever the design is paired, a multiplicity correction and a power statement; intervals are present, the correction and the power statement are not. On automatic metrics, human validation of any LLM-based score and an independent judge, both absent.

Editors will further expect sensitivity analyses over prompt, context size, hybrid weight, dtype and seed, of which only seeds are covered; a Data Availability Statement, CC BY-SA compliance for the Wikipedia-derived text and, per MDPI policy, a declaration of AI-assisted data generation, the last of which is missing; CRediT author contributions, funding and conflict-of-interest statements, all missing; at least four figures (gap by encoder with intervals, the gating curve, the k-sweep and latency) where the manuscript currently has only tables; and a threats-to-validity section covering internal, external and construct validity, the last of which, the judge, is not discussed at all.

5. Minor points

The corrected $n = 200$ run has the oracle condition, in which the gold passage is always supplied, scoring below the MSA-retrieval condition (0.730 versus 0.795); either explain this or do not use that run. The judge-reply parser in the original notebooks silently scores a correct answer as wrong whenever both verdicts land on one line; the corrected notebooks fix this and expose a parse rate, which should be reported (it is 1.000 on the committed run). Report reranker latency together with dtype and hardware, since float32 on a T4 overstates the overhead. Define how the code-switching rate was measured, and unify file naming so that the Data Availability Statement resolves.

6. Concrete revision plan, in the order a jury will weigh it

First, label the 100-row judge sheet with two native speakers blind to the model's verdict, report kappa, raw agreement and per-run kappa, and add one non-Qwen judge on the same sample; this closes A and B in one to two days and without it reviewers stop reading. Second, fix one canonical generation run with a pinned float16 compute dtype, versions, seed and hardware, archive its raw, summary and comparison files on Zenodo, and cite that DOI in every table; this closes C and D in a day. Third, declare Recall@1 as the primary hypothesis, Holm-correct every secondary interval and add a one-paragraph power statement; half a day, closes F and I. Fourth, reword the asymmetry claim as descriptive and rewrite the Related Work to position each contribution as set out in Section 2; a day, closes E and the originality objection. Fifth, run three paraphrased Arabic-only prompts and report the spread per condition; one GPU day, closes G. Sixth, re-select the hybrid weight, rerank depth and context size on a 50-item development split and re-report on the remaining 150; half a day, closes H. Seventh, run a second generator on the same 200 items and obtain two-annotator kappa on Darija authenticity; two to three days, closes J and K.

7. Verdict

A careful, honest study whose retrieval-stage contribution is ready. Its generation-stage numbers come from an unvalidated judge and a pipeline whose output changes materially with a dtype flag. Address A to D, F and the positioning set out in Section 2 before resubmission; E, G and H determine how strongly the conclusions may be worded; I to K set the scope of the generalisation claim. With those in place, the released benchmark and the paper's negative results are its most citable content.